

Practical 5 : Key Exchange using Diffie-Hellman

AIM : Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CODE :

```
import random

p = int(input("Enter prime number p: "))
g = int(input("Enter primitive root g: "))

a = int(input("Enter private key of Alice: "))
b = int(input("Enter private key of Bob: "))

A = pow(g, a, p)
B = pow(g, b, p)

alice_key = pow(B, a, p)
bob_key = pow(A, b, p)

print()
print("Madesh's public key:", A)
print("Manan's public key:", B)

print("Madesh's shared secret:", alice_key)
print("Manan's shared secret:", bob_key)
print()
if alice_key == bob_key:
    print("Key exchange successful.")
else:
    print("Key exchange failed.")
```

OUTPUT :

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/IT-51/Downloads/T120 DHEERAJ SINGH 5th PRACTICAL CYBER SECURITY.py
Enter prime number p: 45
Enter primitive root g: 18
Enter private key of Alice: 12
Enter private key of Bob: 11

Madesh's public key: 36
Manan's public key: 27
Madesh's shared secret: 36
Manan's shared secret: 36

Key exchange successful.
>>>
```